

E-Commerce Sales Analysis and Power BI Dashboard

1. Introduction

This project turns an E-Commerce store's 2025 order data into a four-page Power BI dashboard which the management can easily use. The dashboard shows what sells, what makes a profit, who buys, where they live, and how they pay.

The report explains the data, the business questions, and the four steps used to build the dashboard: cleaning the data, building the data model, creating the measures, and designing the dashboard. It ends with the key findings and the limits of the data.

2. About the Data

I used Claude AI to generate the data. The data comes in two CSV files linked by Order ID. It covers orders placed from January to December 2025 by customers in all eight divisions of Bangladesh.

Table	One row means	Columns
Orders	One complete order	Order ID, Order Date, CustomerName, Division, City
Order Details	One product line in an order	Order ID, Amount, Profit, Quantity, Category, Sub-Category, PaymentMode

One order can have several product lines, so the two tables have a **one-to-many** relationship: one order in Orders matches many lines in Order Details. Products fall into three categories: Electronics, Furniture and Clothing. Customers pay by seven methods: COD, bKash, Nagad, Rocket, Credit Card, Debit Card and EMI.

3. Business Questions

The dashboard answers practical questions in six areas of the business.

Area	Key questions	Main measures
Sales and profit	How much do we sell and earn? Which months are strongest and weakest?	Total Sales, Total Profit, Profit Margin %, MoM Sales Growth %
Products	Which categories and sub-categories sell the most? Which earn the most profit? Which lose money?	Sales, profit and margin by category and sub-category
Customers	Who are the top customers?	Total Customers, Repeat

Area	Key questions	Main measures
	How many buy more than once?	Customers, Sales per Customer
Locations	Which divisions and cities sell the most? Where is profit weak? Do product choices differ by division?	Sales, profit and margin by division and city
Payments	Which payment methods do customers prefer? Which earn the most profit? Do order values differ?	Sales by payment mode, profit by payment mode, Average Order Value
Orders	How many orders do we receive? What is the average order value and size?	Total Orders, Average Order Value, Units per Order

4. Step 1: Clean and Check the Data

I cleaned both tables in Power Query before building anything else, so that every later step works on correct data.

1. **Loaded both files** with Home → Get Data, then opened Power Query with Transform Data.
2. **Fixed the headers.** The Orders table had formatting errors, so I used Use First Row as Headers.
3. **Fixed the Order Date column.** Some dates used "-" as the separator (e.g. 24-01-2025) and others used "/" (e.g. 10/3/2025). I split the column by the delimiter, then created a new column with an if function to rebuild every value as a proper date in day/month/year order.
4. **Set the data types.** Amount and Profit became decimal numbers, Quantity became a whole number, and Order Date became Date only, with no time attached.
5. **Checked the keys.** I confirmed that each Order ID appears only once in Orders, that every order line has a matching order, and that every order has at least one line.
6. **Kept rows with negative profit.** They are real losses, not errors, and they matter for the analysis.

5. Step 2: Build the Data Model

The model uses a **star schema**: one central Sales table surrounded by four small lookup tables. This keeps the report fast, simple and accurate, and each lookup table filters the Sales table.

Four lookup tables filter one central Sales table

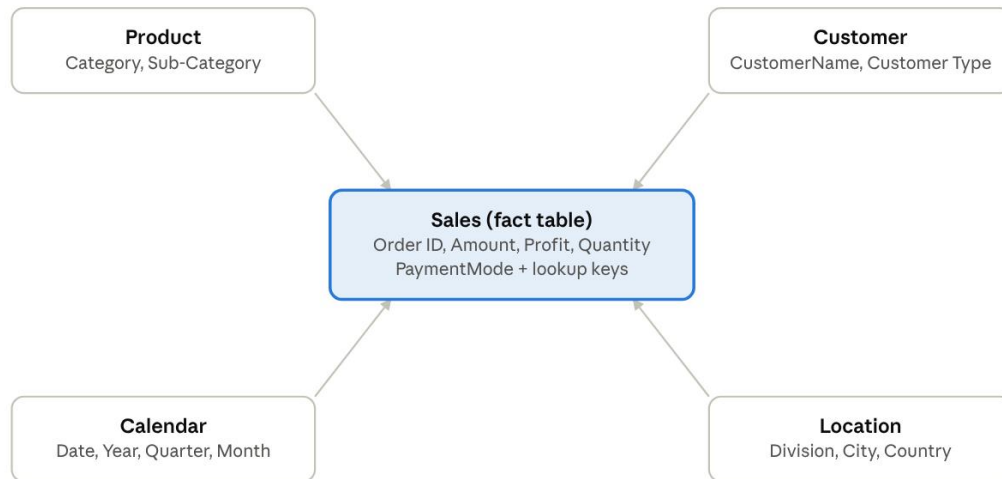


Figure 1: Star schema · 1 fact table, 4 lookup tables

Each arrow is a one-to-many relationship: one row in the lookup table matches many rows in Sales, and filters flow in the arrow's direction.

Merge the tables into Sales

1. I selected Order Details and used Home → Merge Queries → Merge Queries as New, with Orders as the second table.
2. I matched the two tables on Order ID and chose **Left Outer** as the join kind.
3. I expanded the new column to bring in Order Date, CustomerName, Division and City, without the column-name prefix.
4. I renamed the new query **Sales** and checked that it has the same number of rows as Order Details.

Create the lookup tables

Lookup table	Columns kept	Then
Product	Category, Sub-Category	Remove Duplicates
Customer	CustomerName	Remove Duplicates
Location	Division, City	Remove Duplicates; add Country = "Bangladesh"

I turned off Enable load for Orders and Order Details, because they are only staging steps.

Create the date table

I created a Calendar table with DAX covering every day of 2025, with Year, Quarter, Month No, Month and Year-Month columns. I marked it as the date table and sorted Month by Month No so months appear from Jan to Dec.

Connect the tables

In Model view, I linked each lookup table to Sales with a one-to-many relationship and single cross-filter direction:

From (one side)	To (many side)
Calendar[Date]	Sales[Order Date]
Product[Sub-Category]	Sales[Sub-Category]
Customer[CustomerName]	Sales[CustomerName]
Location[City]	Sales[City]

Finally, I hid the duplicate descriptive columns in Sales, so all visuals are built from the lookup tables.

6. Step 3: Create the Key Measures

All calculations were written as DAX measures in a separate Calculation table, so every page uses the same definitions.

Measure	What it shows	How it is calculated
Total Sales	Money received from sales	Sum of Amount
Total Profit	Money earned after costs	Sum of Profit
Profit Margin %	Share of sales kept as profit	Total Profit ÷ Total Sales
Total Orders	Number of separate orders	Distinct count of Order ID
Average Order Value	Average spend per order	Total Sales ÷ Total Orders
Total Quantity	Total units sold	Sum of Quantity
Units per Order	Average items in one order	Total Quantity ÷ Total Orders
Total Customers	Number of different customers	Distinct count of CustomerName
Repeat Customers	Customers with more than one order	Count of customers with 2 or more orders
Repeat Customer %	Share of customers who came back	Repeat Customers ÷ Total Customers
Sales per Customer	Average spend per customer	Total Sales ÷ Total Customers
MoM Sales Growth %	Change from the previous month	(This month – last month) ÷ last month
COD, Credit Card, EMI, Debit	Total sales for each payment	Total Sales filtered to that

Measure	What it shows	How it is calculated
Card, bKash, Nagad, Rocket	mode	payment mode

7. Step 4: Design the Dashboard

The dashboard has four pages, moving from a high-level overview to detail on products, customers and payments.

Common layout

Every page follows the same layout so users always know where to look:

- A header bar with the page title and four slicers: Date, Division, Category and Payment Mode. The slicers are synced, so a selection on one page applies to all pages.
- A row of KPI cards across the top.
- Three or four main visuals below the cards.
- The same colour for each category on every page, and chart titles that state the finding rather than just the topic.

Page 1: Overview

Question it answers: How is the business doing overall?

Visual	Title	Fields
5 KPI cards	Total Sales, Total Profit, Profit Margin %, Total Orders, Average Order Value	One measure per card
Column chart	Sales Peaked in January and Dipped in June–July	Month, Total Sales
Bar chart	Cash on Delivery Brings in the Most Sales	PaymentMode, Total Sales, sorted from highest to lowest
Column chart	Electronics Leads Sales, with Clothing Close Behind	Category, Total Sales
Bar chart	Dhaka and Chattogram Drive Over Half of Sales	Division, Total Sales, top 5 divisions sorted from highest to lowest

Page 2: Products

Question it answers: Which products make money and which lose it?

Visual	Title	Fields
4 KPI cards	Total Sales, Total Profit, Profit Margin %, Total Quantity	One measure per card
Matrix	Printers Earn the Most Profit of Any Sub-Category	Sub-Category; Total Sales, Total Profit, Profit Margin %, Total Quantity, with colour scales, sorted by profit
Scatter chart	High Sales Don't Always Mean High Profit	Total Sales (x), Total Profit (y), one dot per sub-category, coloured by category
Bar chart	Five Sub-Categories Are Losing Money	Sub-Category, Total Profit, filtered to profit below 0 and sorted by largest loss

Page 3: Customers and Locations

Question it answers: Who buys from the platform, and where?

Visual	Title	Fields
4 KPI cards	Total Customers, Repeat Customers, Repeat Customer %, Sales per Customer	One measure per card
Bar chart	Our Top 5 Customers by Sales	CustomerName, Total Sales, using a Top N filter
Donut chart	Only 1 in 3 Customers Buys Again	Customer Type, Total Customers
Treemap	Dhaka and Chattogram Cities Lead Sales	Division (category), City (details), Total Sales; profit and margin in tooltips

I first tried a map for locations, but the Filled Map and Map visuals could not place Bangladeshi cities reliably. The treemap was used instead, because it compares division and city sales clearly without depending on map boundaries.

Page 4: Payments

Question it answers: How do customers pay, and which methods bring the most value?

Visual	Title	Fields
7 KPI cards	COD, Credit Card, EMI, Debit Card, bKash, Nagad, Rocket	Total sales for each payment mode

Visual	Title	Fields
Matrix	How Each Division Pays: COD Leads	Division (rows), PaymentMode (columns), sales shown as percent of row total
Bar chart	EMI Customers Spend the Most per Order	PaymentMode, Average Order Value
Pie chart	COD Makes Up Over a Third of Sales	PaymentMode, Total Sales
Column chart	Credit Card Earns as Much Profit as COD	PaymentMode, Total Profit, sorted from highest to lowest

A monthly payment-share chart was tested first, but with seven payment modes across twelve months it was too crowded to read. The profit-by-payment-mode chart replaced it.

8. Key Findings

The store made **613K Taka in sales and 52K Taka in profit** in 2025, an 8.44% profit margin, from 500 orders with an average value of 1.23K Taka.

Sales by month

- January was the strongest month at 97K Taka, followed by August (74K) and October (68K).
- Sales dropped in the middle of the year: June (30K) and July (29K) were the weakest months.

Products

- **Electronics sells the most but earns the lowest margin.** It brought in 233K Taka of sales at a 7.9% margin. Clothing sold less (202K Taka) but earned the most profit (18.7K Taka) and the highest margin (9.2%). Furniture sold 178K Taka at an 8.2% margin.
- Printers are the most profitable sub-category, with 12,048 Taka profit at a 14.5% margin, followed by Bookcases (9,122 Taka).
- Saree has the highest sales (82,743 Taka) and the most units sold (795), but only a 6.9% margin. Phones sell well (64,566 Taka) but keep just 4.0% as profit.
- T-shirts (20.3%) and Shirts (20.0%) have the best margins, but small sales of about 10K Taka each.
- **Five sub-categories lose money**, 3,205 Taka in total: Furnishings (–1,130), Electronic Games (–899), Three-Piece (–558), Panjabi (–439) and Leggings (–179). Electronic Games stands out, because it made 54,835 Taka of sales and still lost money.

Customers

- 107 of 336 customers (31.85%) ordered more than once; 68% bought only once.

- Each customer spent 1.82K Taka on average. The top customers were Farzana Hossain (13.9K Taka), Jamal Khan (13.1K) and Moumita Hossain (10.9K).

Locations

- **Dhaka (179K Taka) and Chattogram (170K Taka) together bring in 57% of all sales.** Sylhet is a distant third at 66K Taka.
- The top cities are Dhaka (89K Taka), Chattogram (82K), Cumilla (61K) and Moulvibazar (40K). Cumilla and Moulvibazar sell more than several division headquarters.
- Rangpur, Barishal and Mymensingh are among the smallest markets.

Payments

- **Cash on delivery leads with 217K Taka (35.5% of sales).** Cards and EMI together make up about 49%, and mobile wallets (bKash, Nagad, Rocket) about 16%.
- Credit Card earns about the same profit as COD (around 18K Taka each) on only 56% of COD's sales. Nagad and Rocket earn almost no profit.
- EMI has the highest average order value (1,029 Taka), followed by Credit Card (951 Taka). Nagad has the lowest (276 Taka).
- Habits differ by region: COD is strongest in Dhaka (39.4% of sales), Barishal relies most on Credit Card (29.5%), and Mymensingh uses EMI most (25.8%).

9. Limitations

The data shows what happened, but not always why. These limits should be kept in mind when reading the dashboard:

- **Only one year of data.** The data covers 2025 only, so year-over-year comparison is not possible, and January has no previous month for growth.
- **No product names or codes.** Analysis stops at sub-category level.
- **No customer ID.** Customers are matched by name, so two different people with the same name are counted as one.
- **Payment mode is recorded per product line.** One order can use several payment methods, so payment analysis uses share of sales rather than share of orders.
- **Customer type does not change with the date filter.** Repeat and one-time labels are calculated over the whole year.
- **No discounts, shipping costs or returns.** The full cost of each order is not visible.
- **No marketing, website or delivery data.** The dashboard cannot explain why sales rose or fell.
- **No sales targets.** A small target table could be added later to track progress against goals.

10. Conclusion

The project turned two raw CSV files into a clean star-schema model and a four-page Power BI dashboard. Management can now see sales, profit, products, customers, locations and payment methods in one place, and filter every page by date, division, category and payment mode.